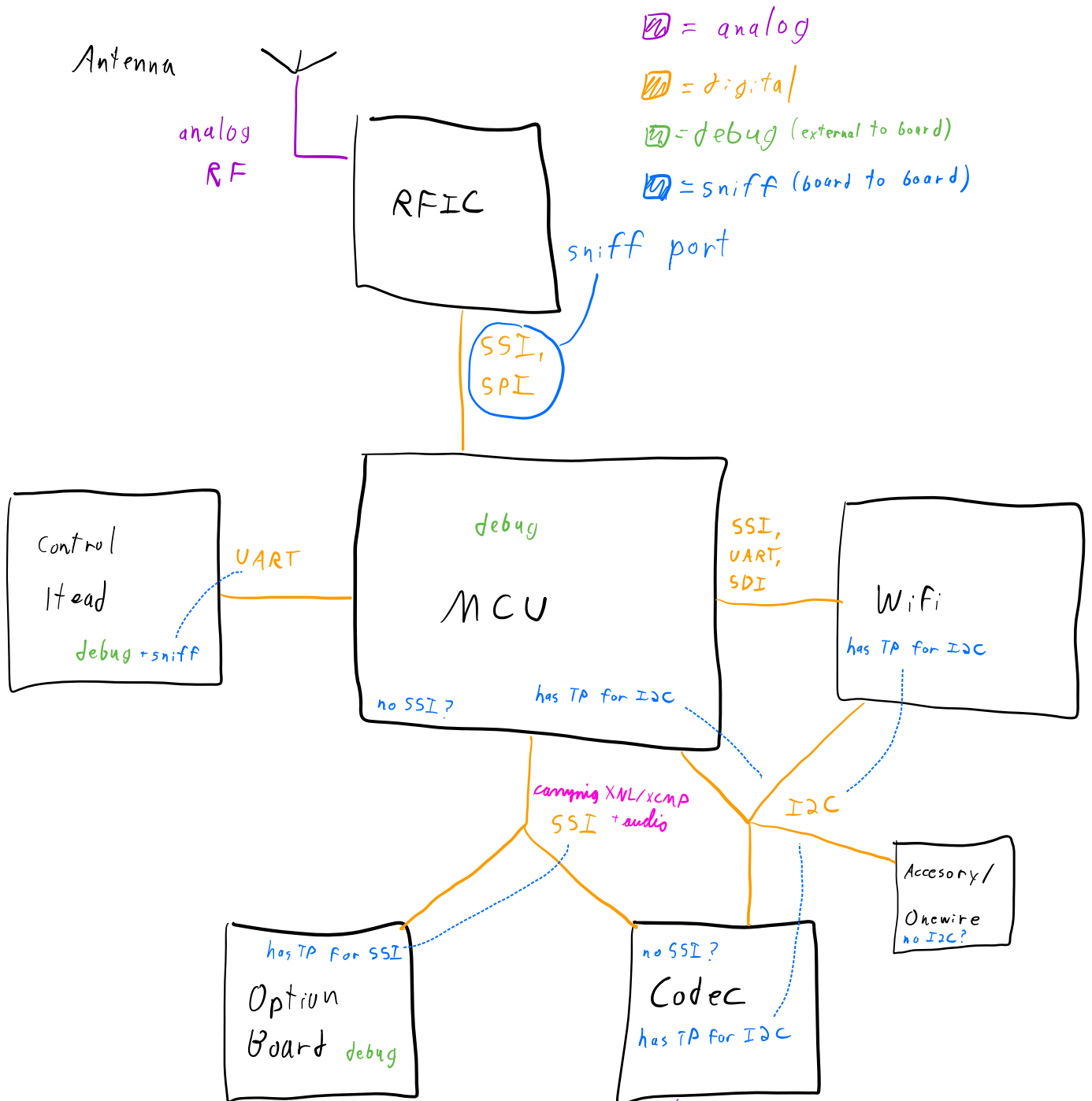
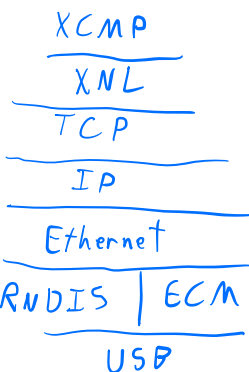


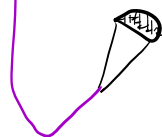


protocol stack



non-IP peripherals use
USB CDC ACM (serial)
instead?

can also turn on some features (e.g. telemetry)
that make radio spit out UDP packets.



Microphone
/ Speaker

Section 1

1.0 MOTOTRBO XPR 5350e/XPR 5550e Mobile Radio Block Diagram

Figure 1-1 shows the MOTOTRBO XPR 5350e/XPR 5550e Mobile Radio Block Diagram. This block diagram should be used for reference for all subsequent chapters.

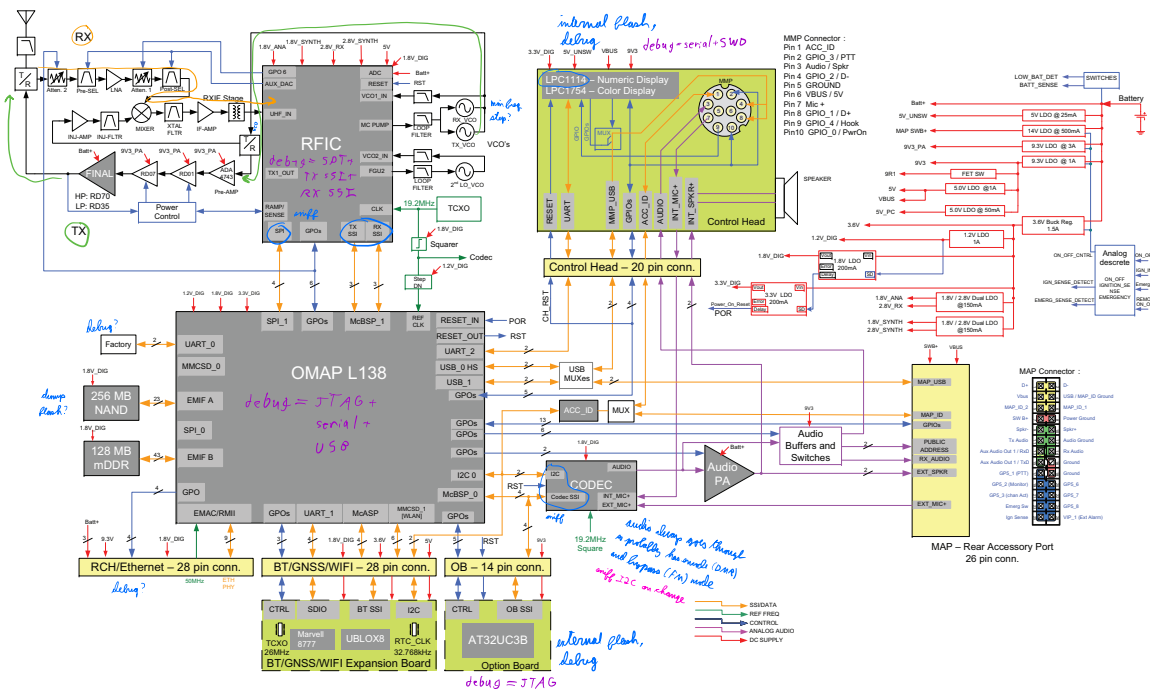


Figure 1-1 Top Level Block Diagram (for PCB PC000942A01-AA)

Debug Headers

are MOLEX slimstack line of mezzanine connectors

RFIC (P0000): 500334-0160 installed. space?

pins 11, 12, 16 are ground

GOB (J5004): 500334-0160 installed. space?

pins 1, 3, 5, 7, 9-14 are ground

MCU (P1000): 500334-0160 installed. space?

pins 11, 12, 16 are ground

Head (P6000): not placed, but footprint for 500334-0160. space?

pins 6-8, 11-16 are ground

pins 11 and 12 ground across all

500334-0160 supposed to be ^{5.2?} ^{2.8?} $5.45 \times 2.92 \text{ mm}$

matching part is 54102-0164 $6.6 \times 4.1 \text{ mm}$

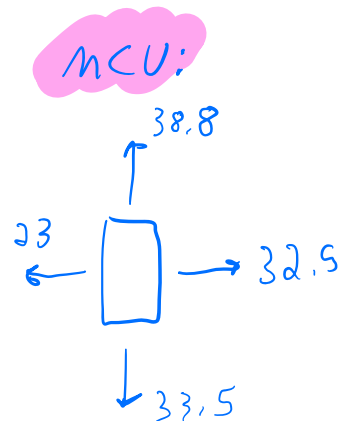
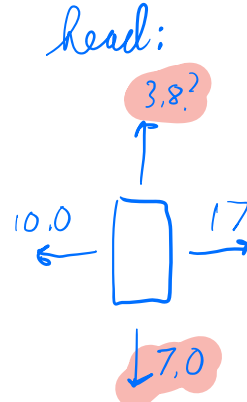
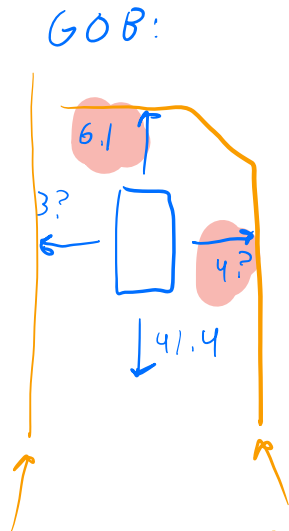
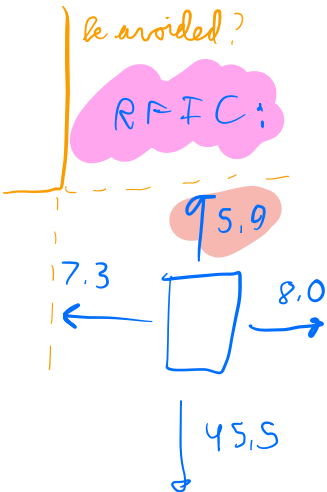
this is 2.5mm stack height

could also be 1.5mm height

55560-0168

male would be 54722-016

meets low enough to
be avoided?



can be avoided
if dismount PCB
from chassis

shield might be
low enough to
avoid? ~3mm



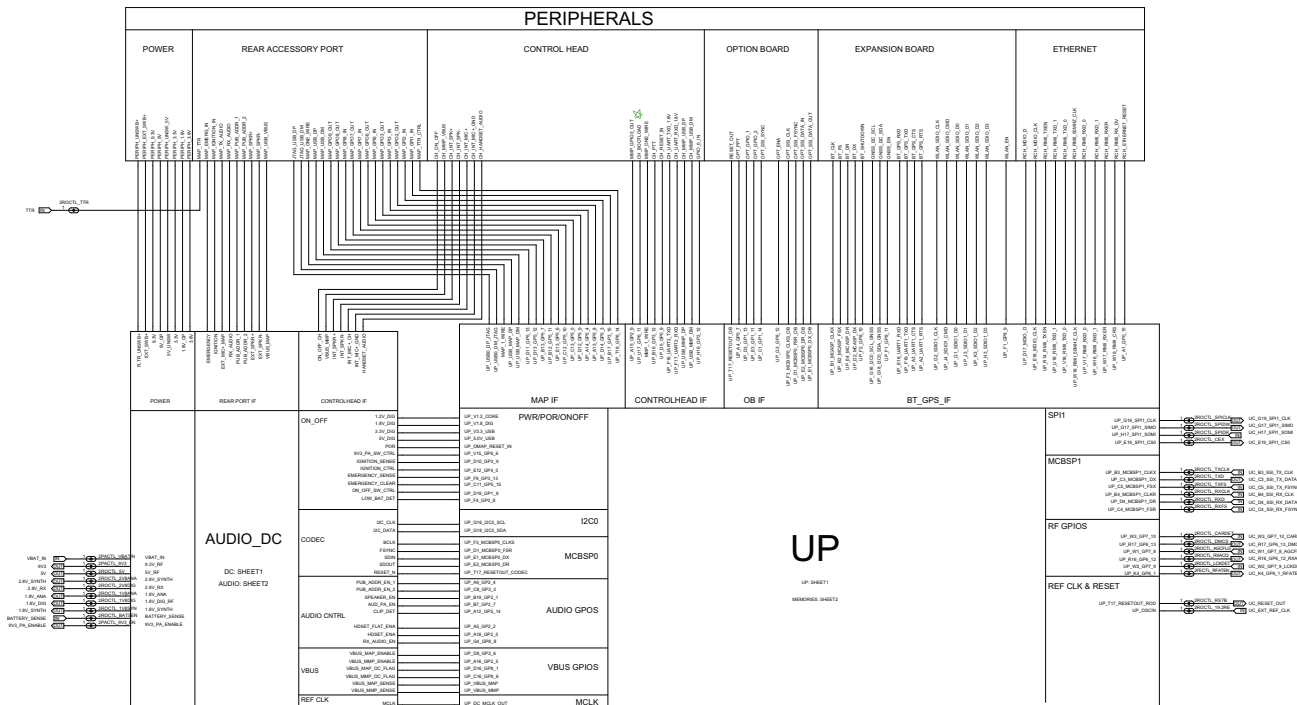
XPR 5350e/XPR 5550e Mobile Radios

Chapter 6

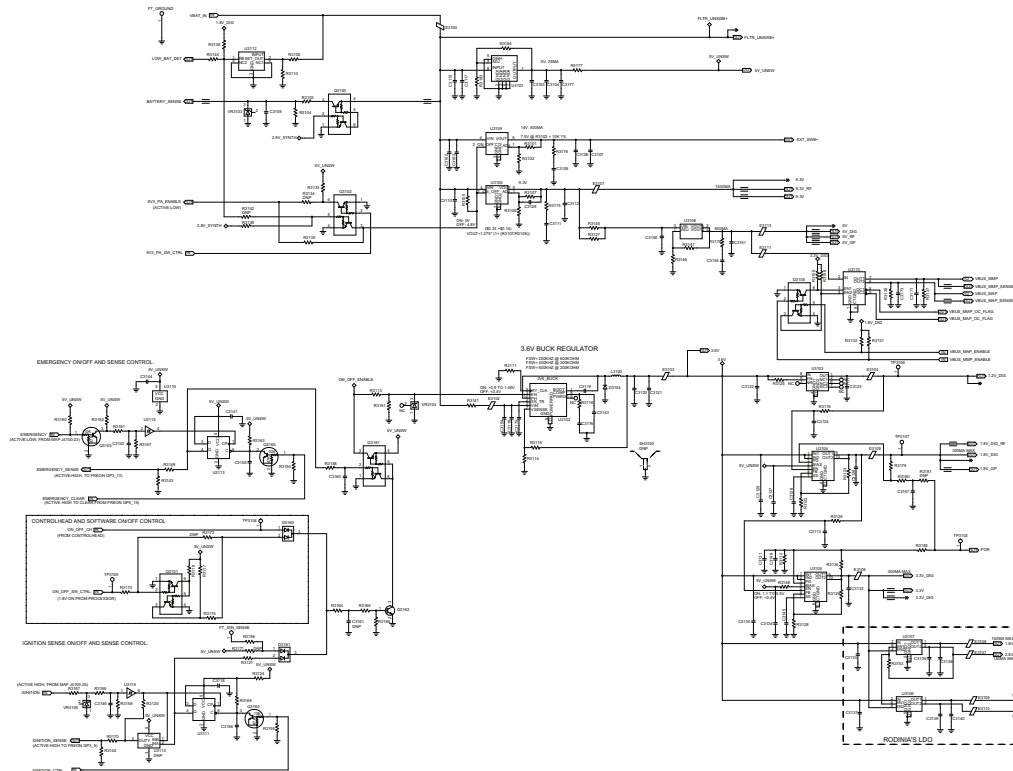
Controller

Service Information

2.0 Controller R3 Schematic Diagrams



Controller – Top Level Schematic Diagram

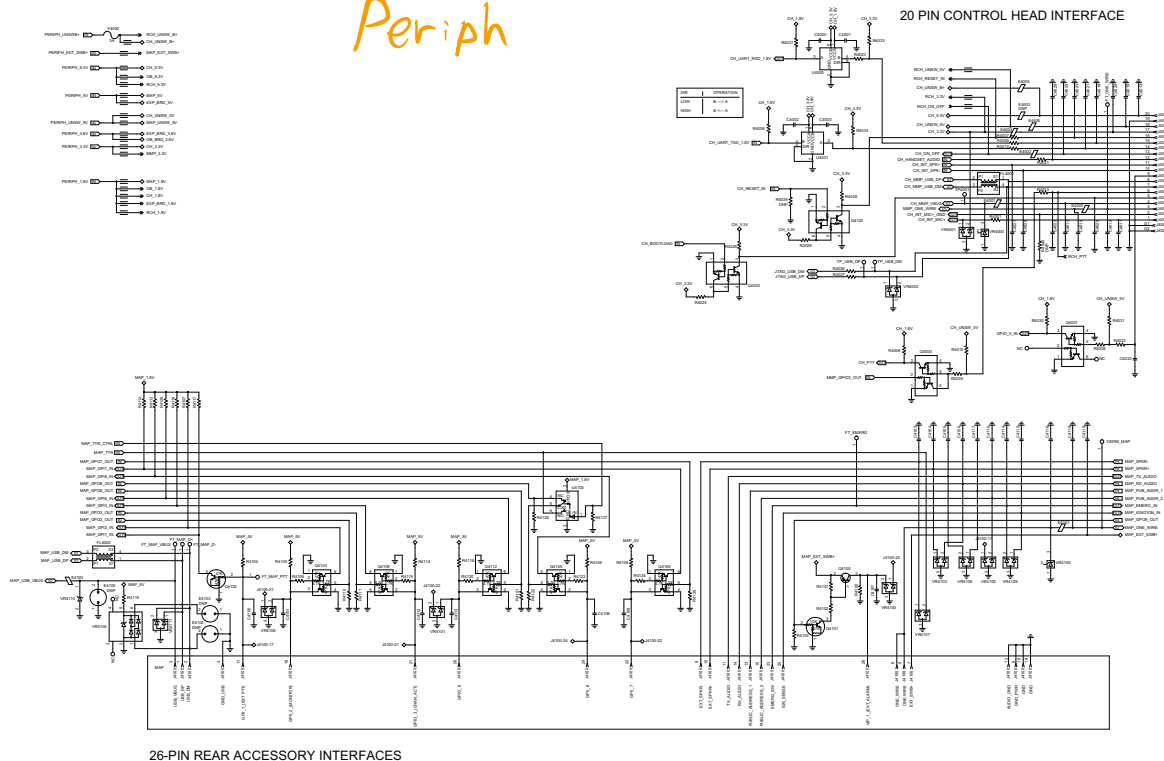


Controller DC Supply Schematic Diagram

[illegible]

Controller Audio Schematic Diagram

Periph

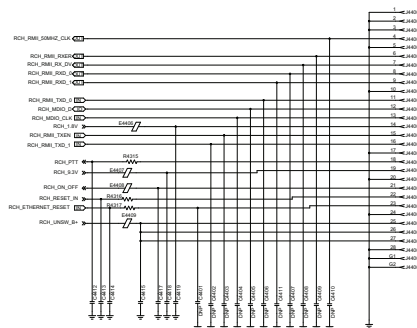


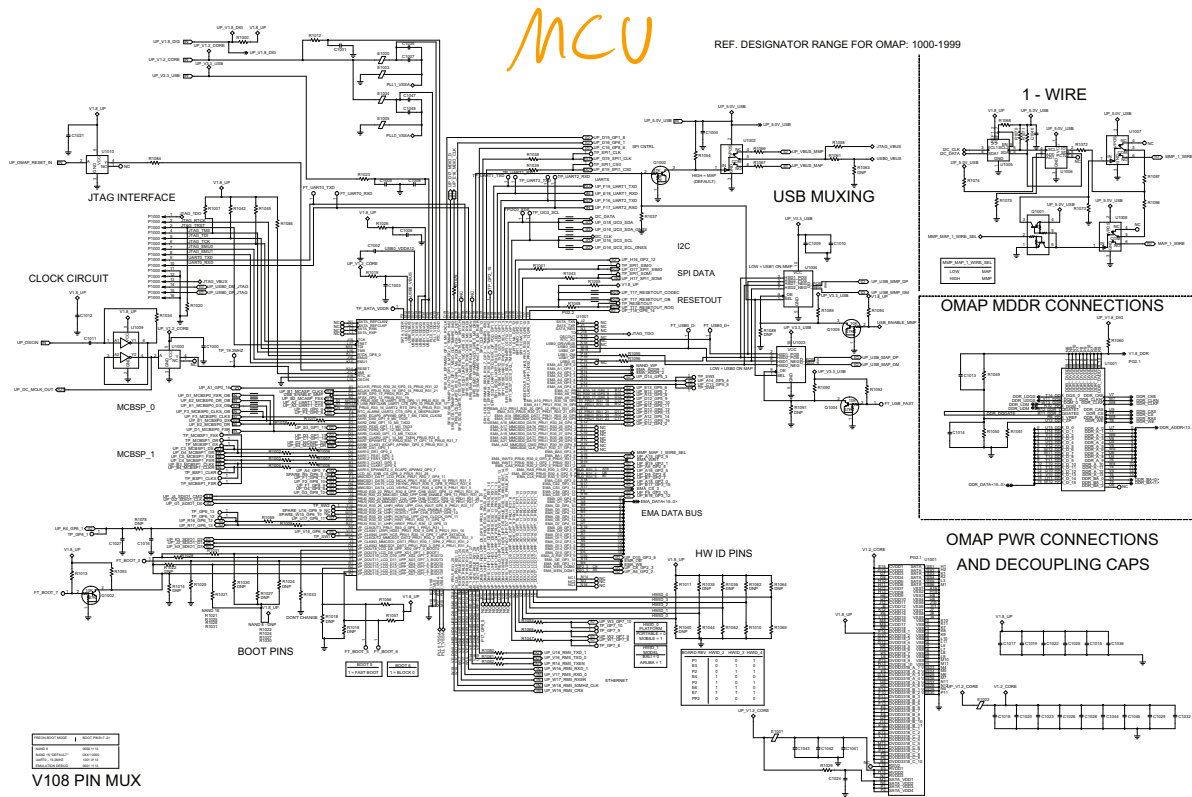
Controller Peripheral Schematic Diagram (sheet 1 of 2)



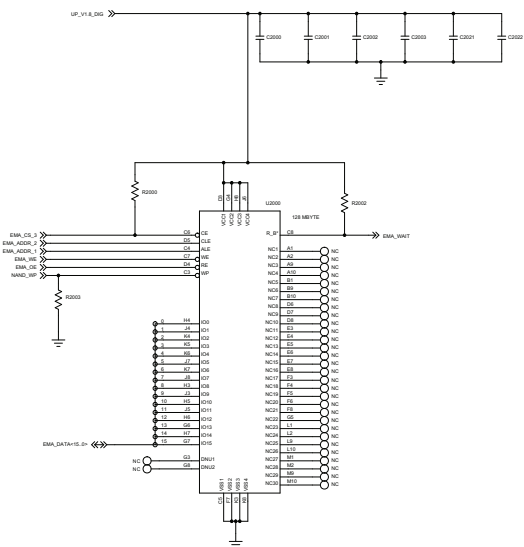
The schematic diagram illustrates the PCB layout for the ESP8266 module. It shows the placement of various components and the routing of signals and power lines. The components and their values are listed in the table on the right:

Component	Value	Pin
ESP8266	ESP8266	1
ESP8266	ESP8266	2
ESP8266	ESP8266	3
ESP8266	ESP8266	4
ESP8266	ESP8266	5
ESP8266	ESP8266	6
ESP8266	ESP8266	7
ESP8266	ESP8266	8
ESP8266	ESP8266	9
ESP8266	ESP8266	10
ESP8266	ESP8266	11
ESP8266	ESP8266	12
ESP8266	ESP8266	13
ESP8266	ESP8266	14
ESP8266	ESP8266	15
ESP8266	ESP8266	16
ESP8266	ESP8266	17
ESP8266	ESP8266	18
ESP8266	ESP8266	19
ESP8266	ESP8266	20
ESP8266	ESP8266	21
ESP8266	ESP8266	22
ESP8266	ESP8266	23
ESP8266	ESP8266	24
ESP8266	ESP8266	25
ESP8266	ESP8266	26
ESP8266	ESP8266	27
ESP8266	ESP8266	28
ESP8266	ESP8266	29
ESP8266	ESP8266	30
ESP8266	ESP8266	31
ESP8266	ESP8266	32
ESP8266	ESP8266	33
ESP8266	ESP8266	34
ESP8266	ESP8266	35
ESP8266	ESP8266	36
ESP8266	ESP8266	37
ESP8266	ESP8266	38
ESP8266	ESP8266	39
ESP8266	ESP8266	40
ESP8266	ESP8266	41
ESP8266	ESP8266	42
ESP8266	ESP8266	43
ESP8266	ESP8266	44
ESP8266	ESP8266	45
ESP8266	ESP8266	46
ESP8266	ESP8266	47
ESP8266	ESP8266	48
ESP8266	ESP8266	49
ESP8266	ESP8266	50
ESP8266	ESP8266	51
ESP8266	ESP8266	52
ESP8266	ESP8266	53
ESP8266	ESP8266	54
ESP8266	ESP8266	55
ESP8266	ESP8266	56
ESP8266	ESP8266	57
ESP8266	ESP8266	58
ESP8266	ESP8266	59
ESP8266	ESP8266	60
ESP8266	ESP8266	61
ESP8266	ESP8266	62
ESP8266	ESP8266	63
ESP8266	ESP8266	64
ESP8266	ESP8266	65
ESP8266	ESP8266	66
ESP8266	ESP8266	67
ESP8266	ESP8266	68
ESP8266	ESP8266	69
ESP8266	ESP8266	70
ESP8266	ESP8266	71
ESP8266	ESP8266	72
ESP8266	ESP8266	73
ESP8266	ESP8266	74
ESP8266	ESP8266	75
ESP8266	ESP8266	76
ESP8266	ESP8266	77
ESP8266	ESP8266	78
ESP8266	ESP8266	79
ESP8266	ESP8266	80
ESP8266	ESP8266	81
ESP8266	ESP8266	82
ESP8266	ESP8266	83
ESP8266	ESP8266	84
ESP8266	ESP8266	85
ESP8266	ESP8266	86
ESP8266	ESP8266	87
ESP8266	ESP8266	88
ESP8266	ESP8266	89
ESP8266	ESP8266	90
ESP8266	ESP8266	91
ESP8266	ESP8266	92
ESP8266	ESP8266	93
ESP8266	ESP8266	94
ESP8266	ESP8266	95
ESP8266	ESP8266	96
ESP8266	ESP8266	97
ESP8266	ESP8266	98
ESP8266	ESP8266	99
ESP8266	ESP8266	100

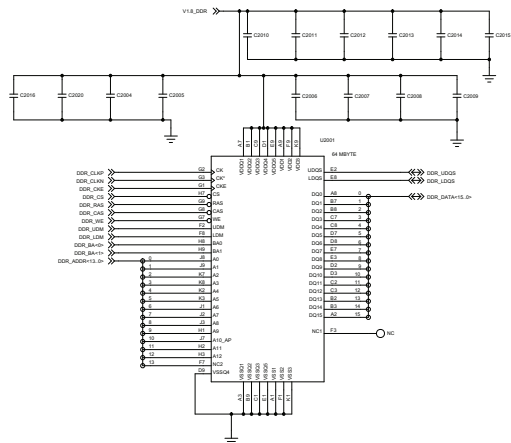




Controller Microprocessor Schematic Diagram



MDDR





XPR[™] 5380e/XPR[™] 5580e

Mobile Radios

Chapter 12

800/900M (806–941MHz)

10–35W/30W

Service Information

2.0 800/900 MHz (806–941 MHz) 10–35/30W PCB/Schematics/Parts List

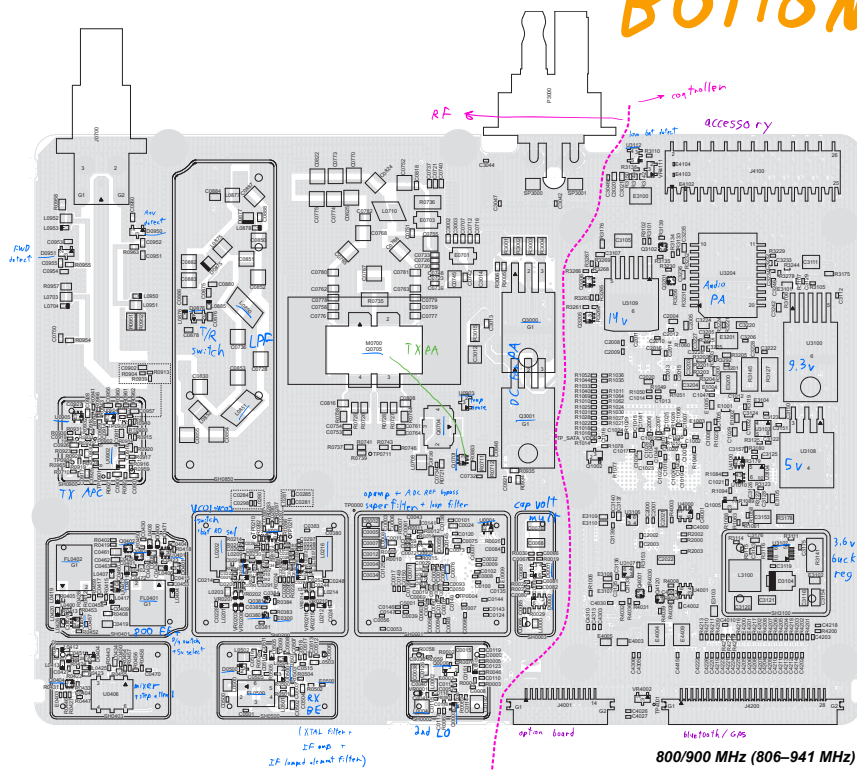
BOTTOM

in shields:

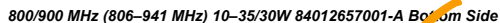
- RX PE
- RX BE
- VCO
- TX APC
- RFIC

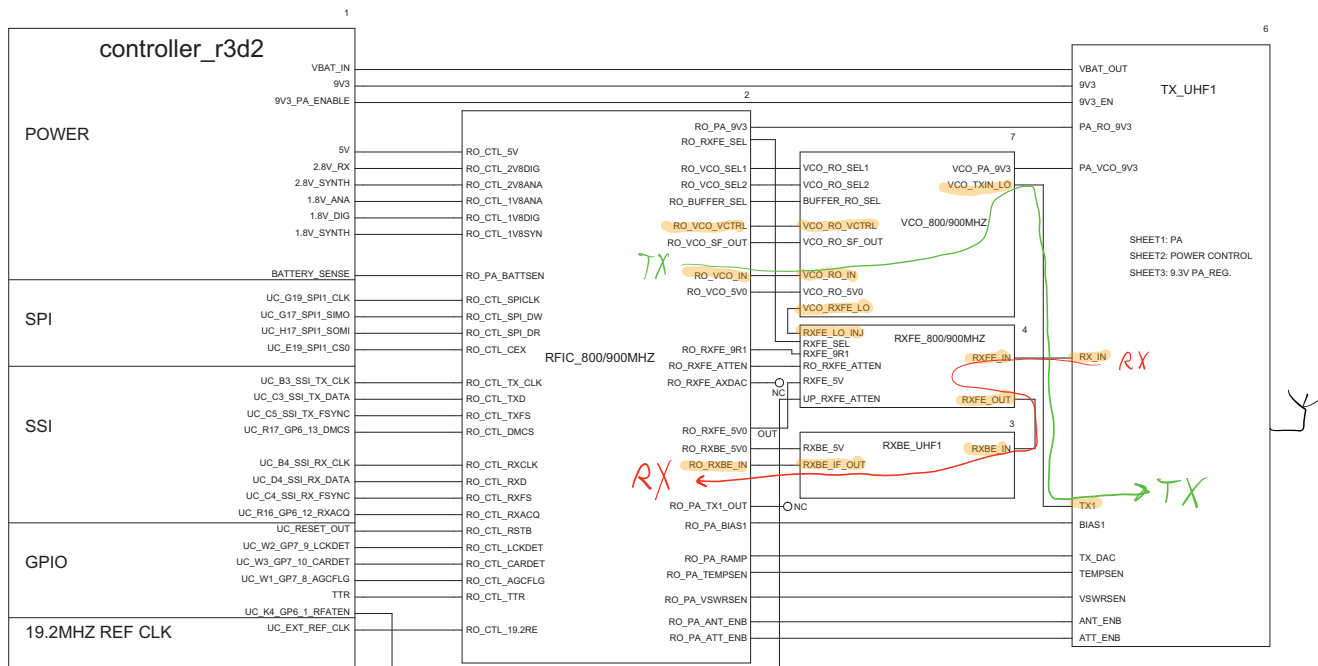
out of shields:

- TX PA
- power

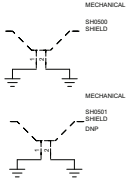
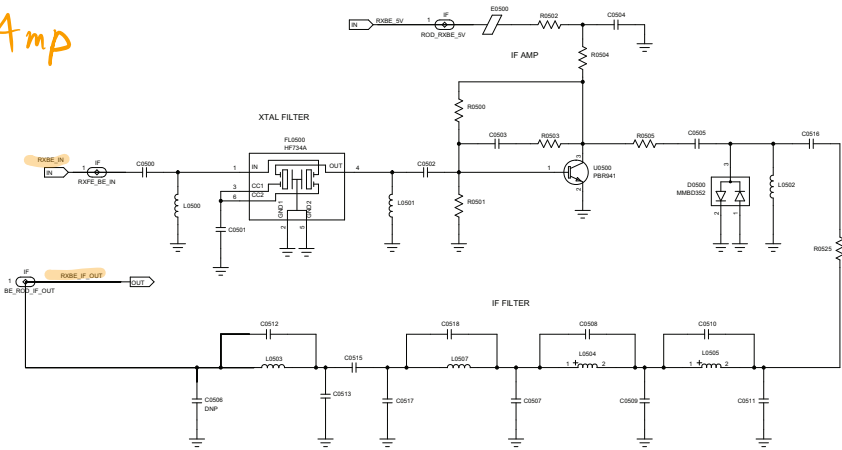


800/900 MHz (806–941 MHz) 10–35/30W 84012657001-A Top Side



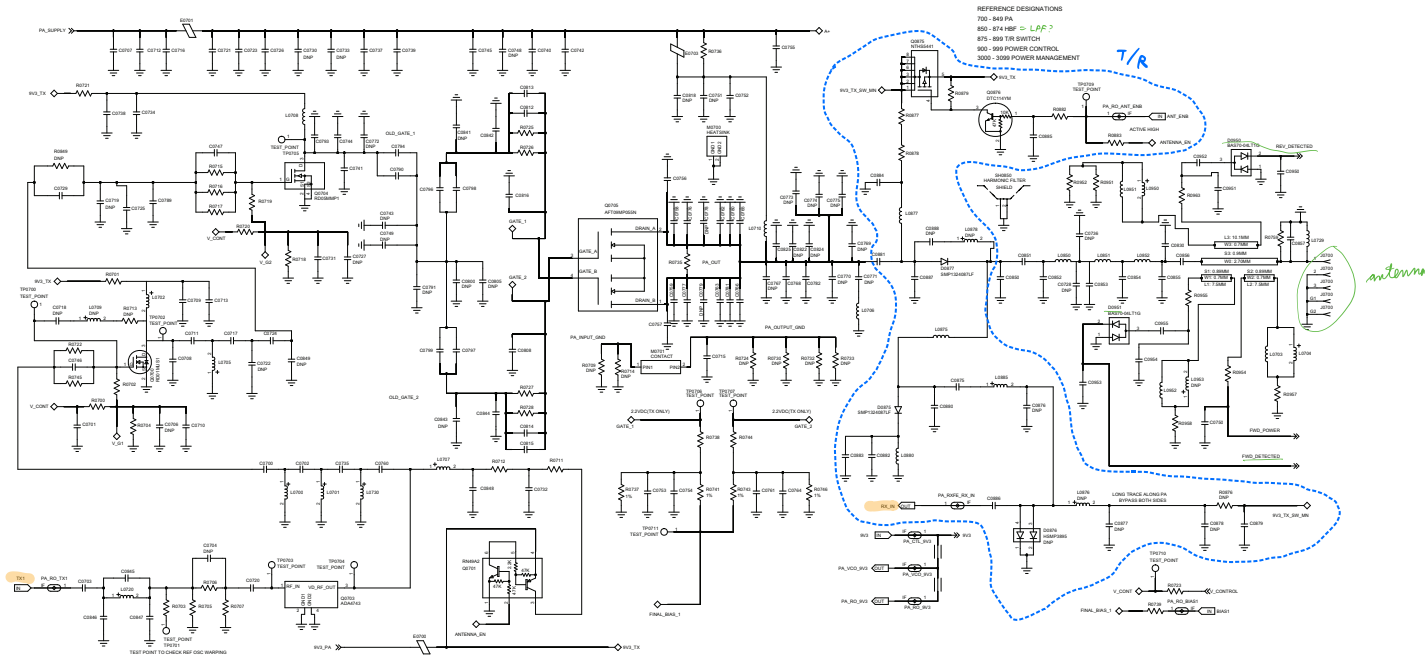


RX IF Amp



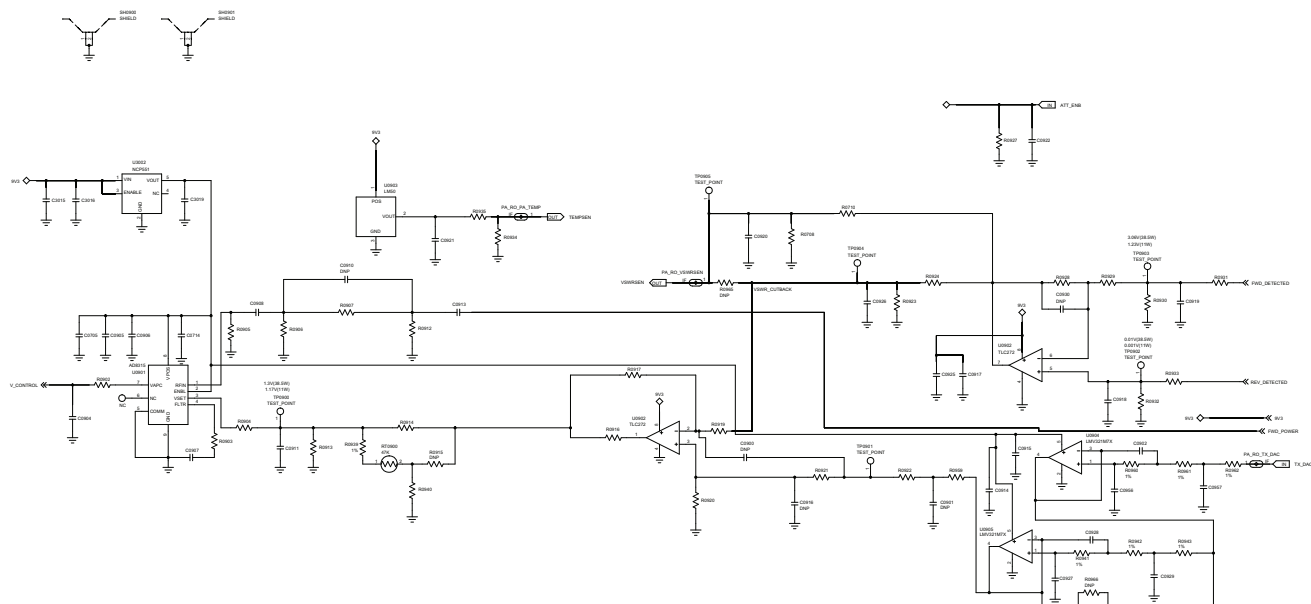
800/900 MHz (806–941 MHz) 10–35/30W Receiver IF Amp Schematic Diagram

Tx PA



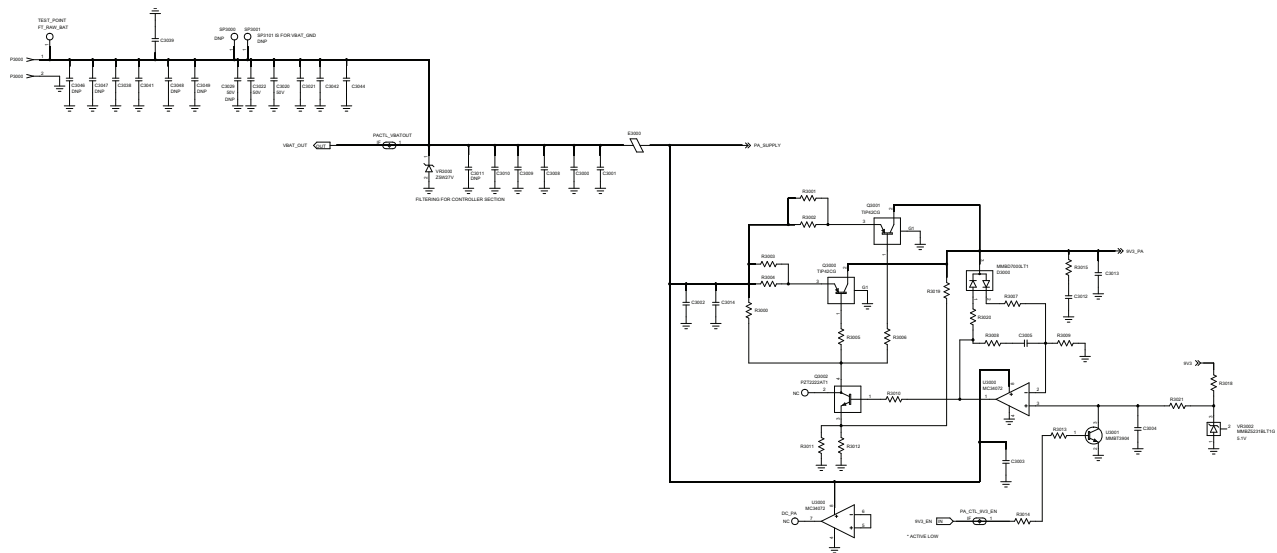
800/900 MHz (806–941 MHz) 10–35/30W Transmitter Power Amplifier Schematic Diagram

TX APC

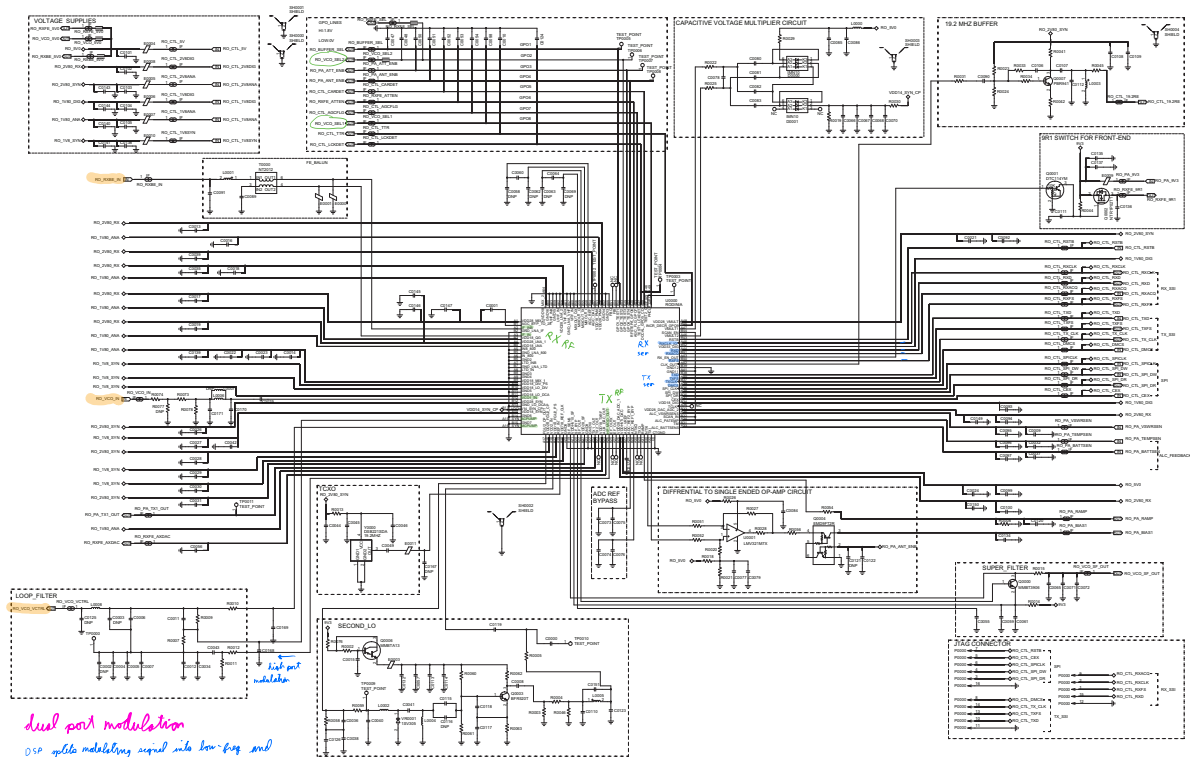


800/900 MHz (806-941 MHz) 10-35/30W Power Control Schematic Diagram

Power



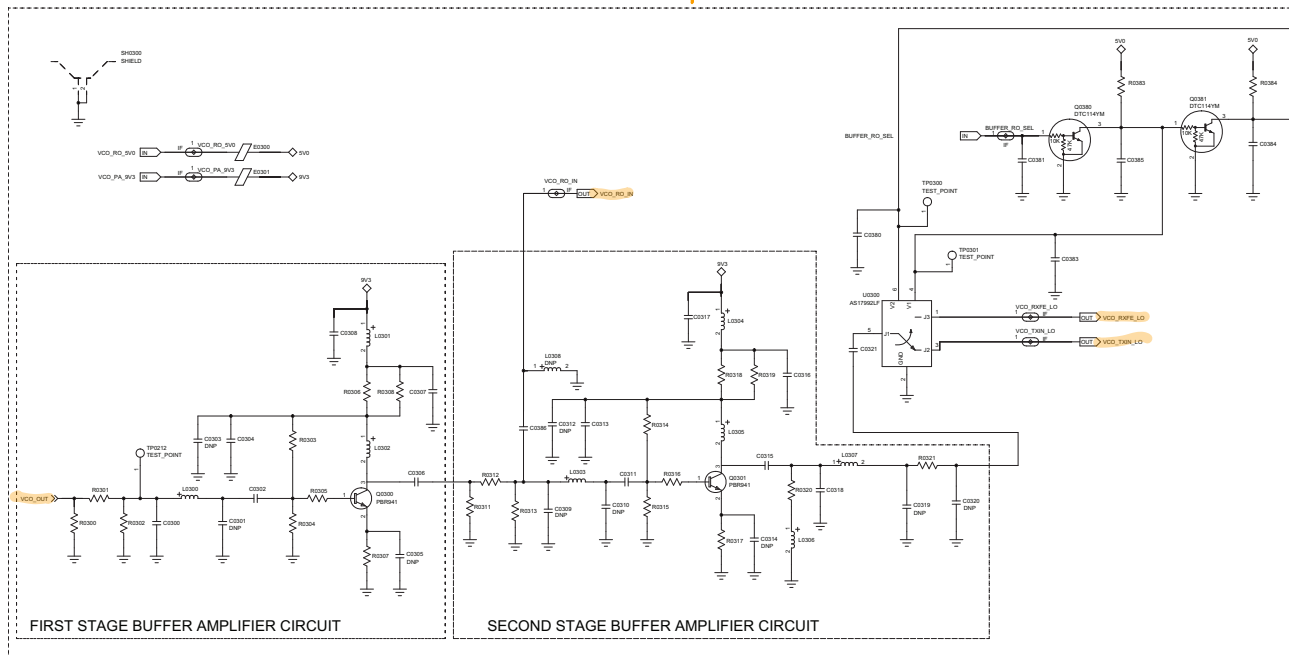
800/900 MHz (806–941 MHz) 10–35/30W Transmitter Power Amplifier DC Distribution Schematic Diagram



800/900 MHz (806-941 MHz) 10-35/30W RFIC Schematic Diagram

verbal description in APX5000/APX6000/SER2000 detailed service manual

VCO Amp



800/900 MHz (806–941 MHz) 10–35/30W VCO Buffer Schematic Diagram



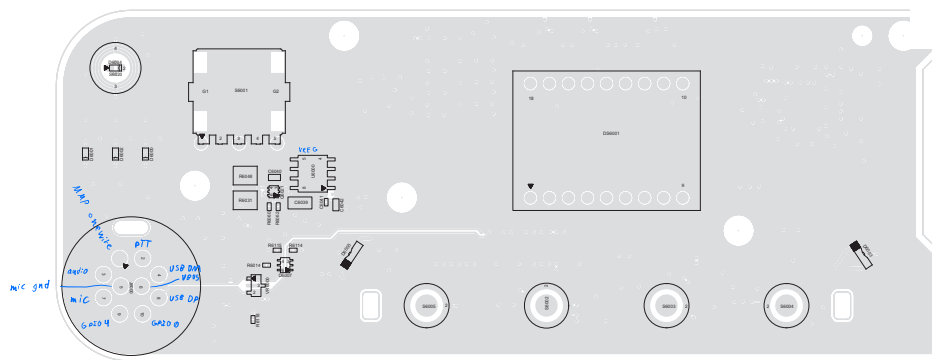
XPR 5350e/XPR 5550e Mobile Radios

Chapter 3

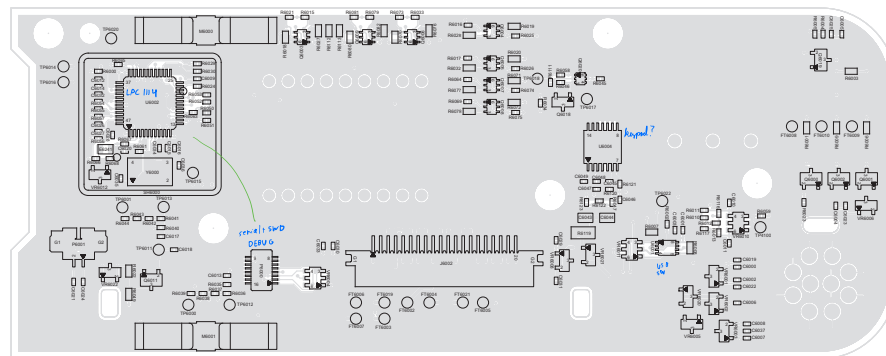
Control Head

Service Information

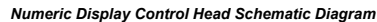
2.0 Numeric Display Control Head – PCB 84012212002-B/Schematics



Top Side View



Bottom Side View



3.0 Numeric Display Control Head Parts List (Rev. B)

Circuit Ref.	Motorola Part No.
C6000	2113945A05
C6001	2113945B02
C6002	2113945A05
C6003	2113946B04
C6004	2113945A05
C6005	2113945B02
C6006	2113945A05
C6007	2113945A05
C6008	2113945A05
C6009	2113945B02
C6010	2113945B02
C6011	2113945B02
C6012	2113946B04
C6013	2113945B02
C6014	2113945B02
C6015	2113946B04
C6016	2113945B02
C6017	2113945B02
C6018	2113946B04
C6019	2113945A05
C6020	2113945A05
C6021	2113945A05
C6022	2113945A05
C6023	2113945A05
C6024	2113945A05
C6025	2113945A05
C6026	NOT PLACED
C6027	NOT PLACED
C6028	2113945A05
C6029	2113945A05
C6030	2113945A05
C6031	2113945A05
C6032	NOT PLACED
C6033	2113956A51
C6034	NOT PLACED

Circuit Ref.	Motorola Part No.
C6035	NOT PLACED
C6036	NOT PLACED
C6037	2113945A05
C6039	2113955D13
C6040	2113944C06
C6041	2113944A44
C6042	2113946D05
C6043	2113956C37
C6044	2113956C37
C6045	2113945A05
C6046	2113945A05
C6047	2113945B02
C6048	2113946B04
C6049	2113945A05
D6000	4815809H04
D6001	4815809H02
D6002	4815809H03
D6003	48012173001
D6004	4815809H04
D6005	48012173001
D6007	48012049001
DS6001	7216958H01
E6241	NOT PLACED
J6000	0964501H04
J6002	0915767H01
M6000	39012078001
M6001	39012078001
P6000	2887818K02
P6001	2815162H01
Q6000	4815261H01
Q6001	4815261H01
Q6002	4815261H01
Q6003	48012232001
Q6004	48012232001
Q6005	48012232001
Q6006	48012232001
Q6007	48012232001
Q6008	48012232001

Circuit Ref.	Motorola Part No.
Q6009	48012232001
Q6011	4815261H01
Q6018	4885065Y01
Q6019	4815261H01
Q6020	4809939C47
Q6021	4815272H01
R6000	0613952R66
R6001	0613952H47
R6002	0613952Q85
R6003	0613952H47
R6004	0613952Q85
R6005	0613952Q93
R6006	0613952H47
R6007	0613952J09
R6008	0613952J09
R6009	0613952H47
R6010	0613952R01
R6011	NOT PLACED
R6012	0613952R01
R6013	NOT PLACED
R6014	0613952R01
R6015	0613952R01
R6016	0613952R01
R6017	0613952R01
R6018	0613952H44
R6019	0613952H44
R6020	0613952H44
R6021	0613952R01
R6022	0613952C01
R6023	0613952R01
R6024	0613952R01
R6025	0613952R01
R6026	0613952R01
R6027	0613952H44
R6028	0613952H44
R6029	0613952R01
R6030	NOT PLACED
R6031	0613959G57

Circuit Ref.	Motorola Part No.
R6032	0613952H44
R6033	0613952R01
R6034	NOT PLACED
R6036	0613952R04
R6037	0613952R09
R6038	0613952R17
R6039	NOT PLACED
R6040	0613952R18
R6041	0613952R04
R6042	0613952R09
R6043	NOT PLACED
R6044	NOT PLACED
R6045	0613952Q89
R6046	0613952Q89
R6047	0613952R18
R6048	0613959G57
R6049	0613952C01
R6050	NOT PLACED
R6051	0613952R66
R6052	0613952N01
R6053	0613952Z44
R6054	0613952R66
R6055	NOT PLACED
R6056	0613952R66
R6057	0613952R66
R6058	0613952Q25
R6059	0613952R01
R6060	0613952R01
R6061	0613952R66
R6062	0613952Q85
R6063	0613952R09
R6064	0613952R01
R6065	0613952R66
R6066	0613952Q73
R6067	0613952R17
R6068	0613952R66
R6069	0613952R01
R6070	0613952H44

Circuit Ref.	Motorola Part No.
R6071	0613952H44
R6072	0613952H44
R6073	0613952R01
R6074	0613952R01
R6075	0613952R01
R6076	0613952H44
R6077	0613952H44
R6078	0613952H44
R6079	0613952R01
R6080	0613952H44
R6081	0613952R01
R6082	0613952H44
R6111	0613952R01
R6112	0613952G67
R6113	0613952G67
R6114	0613952Q73
R6115	0613952Q73
R6116	0613952R66
R6117	0613952R66
R6118	NOT PLACED
R6119	0615871H01
R6120	0613952R43
R6121	0613952R01
R6122	0613952R49
R6123	0613952Q73
S6001	40012062001
S6002	NOT PLACED
S6003	NOT PLACED
S6004	NOT PLACED
S6005	NOT PLACED
S6020	NOT PLACED
SH6000	NOT PLACED
U6000	5114014A03
U6001	51012327001
U6002	51012145002
U6004	5186691V50
VR6000	48012153001
VR6001	4813977A43

Circuit Ref.	Motorola Part No.
VR6002	4813977A43
VR6003	4813977A43
VR6004	4813979P10
VR6005	NOT PLACED
VR6006	4813977A43
VR6008	4813977A50
VR6010	4813979P10
VR6011	4813979P10
VR6012	4813977A43
VR6020	NOT PLACED
VR6021	4813977A43
VR6022	4813977A43
Y6000	NOT PLACED

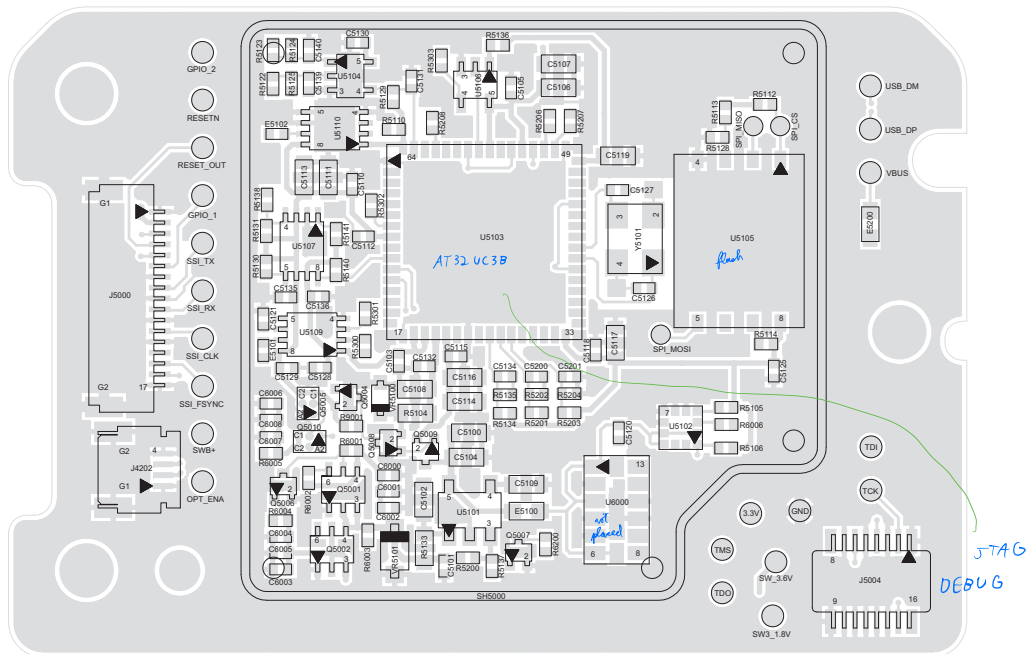


XPR 5350e/XPR 5550e Mobile Radios

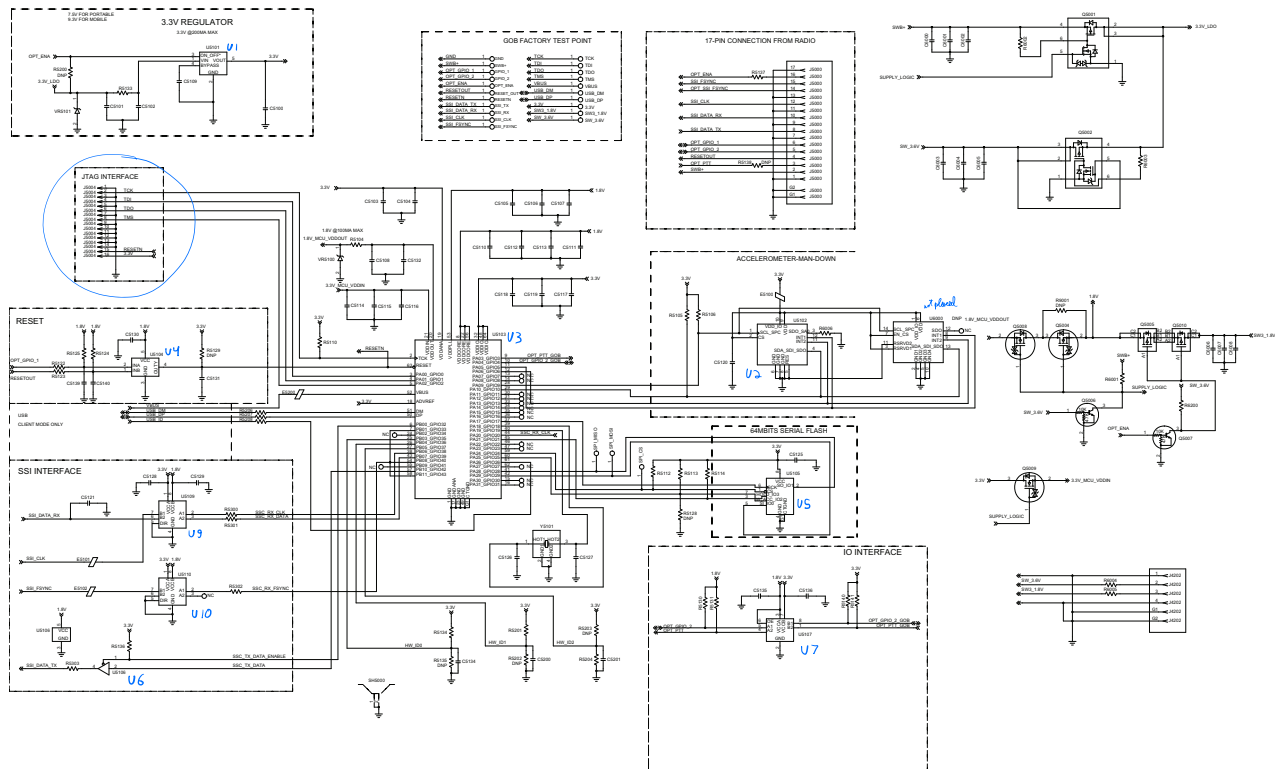
Chapter 5

Generic Option Board (GOB)

Service Information



GOB – PCB PC000957A01-CA Bottom Side



GOB Schematic Diagram

3.0 GOB Parts List

Circuit Ref.	Motorola Part No.
C5100	2113946S35
C5101	2113945A05
C5102	2113946S35
C5103	2113945Y02
C5104	2113945D01
C5105	2113945A11
C5106	2113945D01
C5107	2113956B21
C5108	2113956B21
C5109	2113741F49
C5110	2113945A11
C5111	2113945D01
C5112	2113945Y02
C5113	2113956B21
C5114	2113956B21
C5115	2113945Y02
C5116	2113945D01
C5117	2113945D01
C5118	2113945Y02
C5119	2113956B21
C5120	2113946B04
C5121	2113944A46
C5125	2113945Y02
C5126	2115153H22
C5127	2115153H22
C5128	2113945Y02
C5129	2113945Y02
C5130	2113945Y02
C5131	2113945Y02

Circuit Ref.	Motorola Part No.
C5132	2113945A05
C5134	2113945Y02
C5135	2113945Y02
C5136	2113945Y02
C5139	2113945Y02
C5140	2113945Y02
C5200	2113945Y02
C5201	2113945Y02
C6000	21667009156
C6001	21667009166
C6002	2113945A11
C6003	21667009156
C6004	21667009166
C6005	2113945A11
C6006	2113945A11
C6007	21667009166
C6008	21667009156
E5100	7686949J08
E5101	530004-001-00
E5102	530004-001-00
E5200	24001720002
J4202	CN000225A01
J5000	09012068001
J5004	2887818K02
Q5001	48012274001
Q5002	4888695V01
Q5004	48012088001
Q5005	4870370A72
Q5006	4816134H01
Q5007	4816134H01

Circuit Ref.	Motorola Part No.
Q5008	48012088001
Q5009	48012088001
Q5010	4870370A72
R5104	0613952G67
R5105	0613952R01
R5106	0613952R01
R5110	0613952R01
R5112	0613952R25
R5113	0613952R25
R5114	0613952R25
R5122	0613952Q25
R5123	0613952Q25
R5124	0613952R01
R5125	0613952R01
R5128	NOT PLACED
R5129	NOT PLACED
R5130	0613952R01
R5131	0613952R01
R5133	0613952G67
R5134	0613952R01
R5135	NOT PLACED
R5136	0613952R01
R5137	0613952R66
R5138	NOT PLACED
R5140	0613952R01
R5141	0613952R01
R5200	NOT PLACED
R5201	0613952R01
R5202	NOT PLACED
R5203	NOT PLACED

Circuit Ref.	Motorola Part No.
R5204	0613952R01
R5206	0613952K58
R5207	0613952K58
R5208	0613952R01
R5300	0613952Q33
R5301	0613952Q33
R5302	0613952Q33
R5303	0613952Q73
R6001	0613952R01
R6002	0613952P01
R6003	0613952P01
R6004	0613952R66
R6005	0613952R66
R6006	0613952R01
R6200	0613952R01
R9001	NOT PLACED
SH5000	26012099001
U5101	5171417N02
U5102	IC000643A01
U5103	51012009003
U5104	51012266001
U5105	51012336001
U5106	5188085K03
U5107	51012205001
U5109	5171767H01
U5110	5171767H01
U6000	NOT PLACED
VR5100	48012131001
VR5101	4815040H01
Y5101	93012000001

Circuit Ref.	Motorola Part No.
PCB	PC000957A01

same

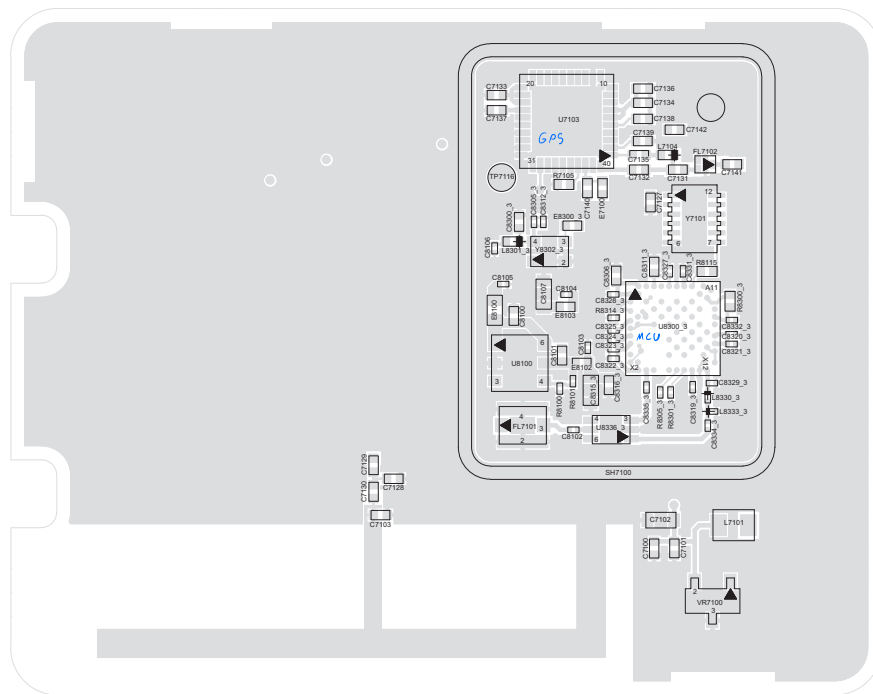


XPR 5350e/XPR 5550e Mobile Radios

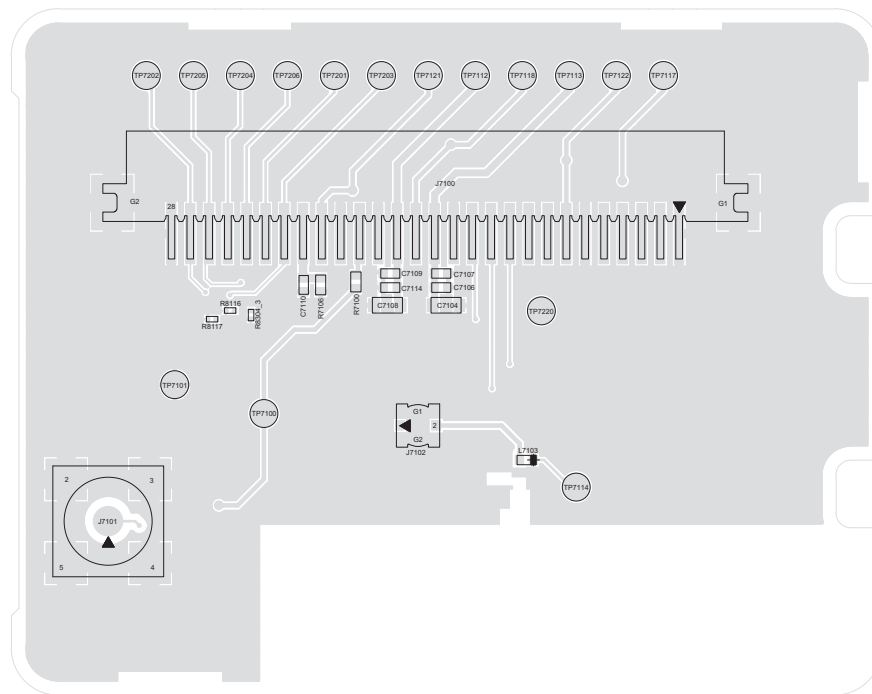
Chapter 4

Bluetooth[®]/Wi-Fi/GNSS Expansion Board Service Information

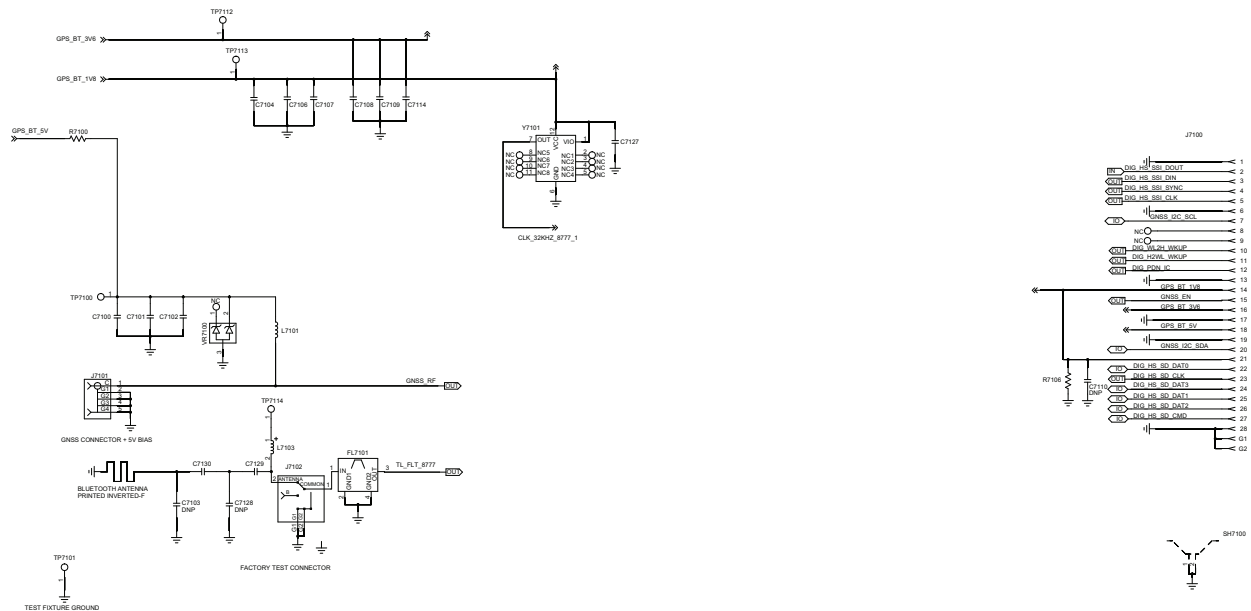
2.0 Bluetooth/Wi-Fi/GNSS (PMLN7252_S) Expansion Board – PCB PC000942A01-AA/Schematic



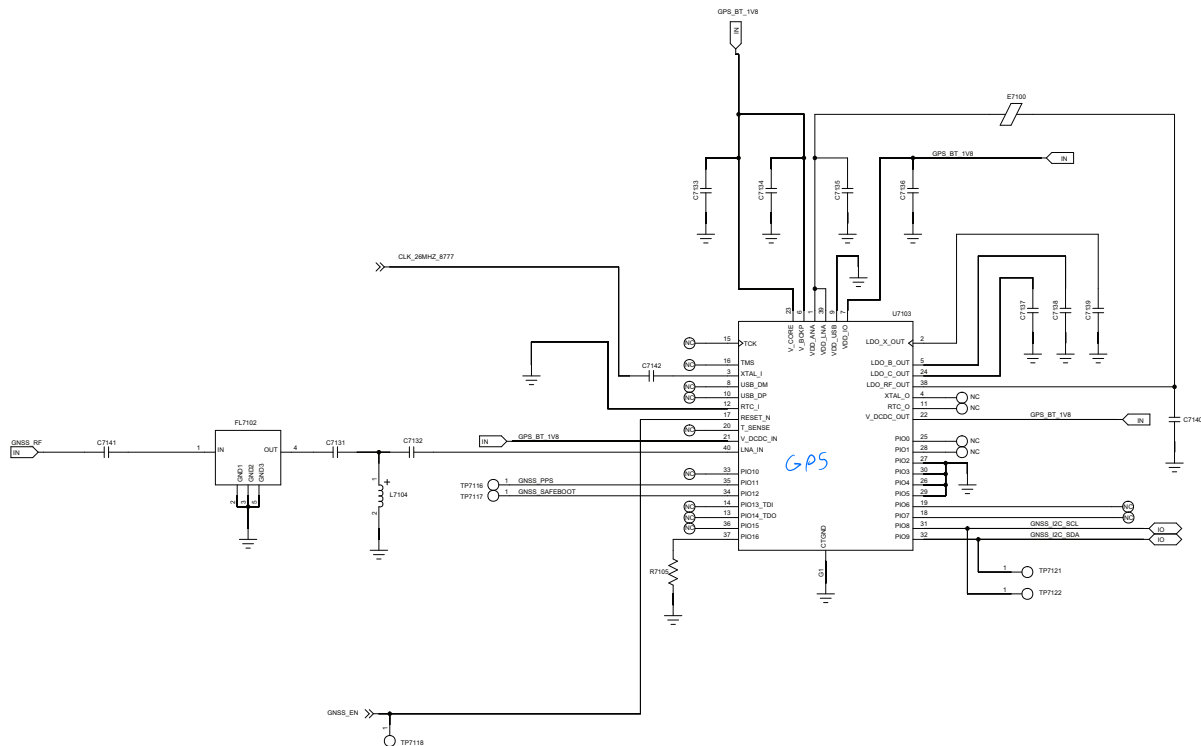
Bluetooth/Wi-Fi/GNSS (PMLN7252_S) Expansion Board – PCB PC000942A01-AA Top Side



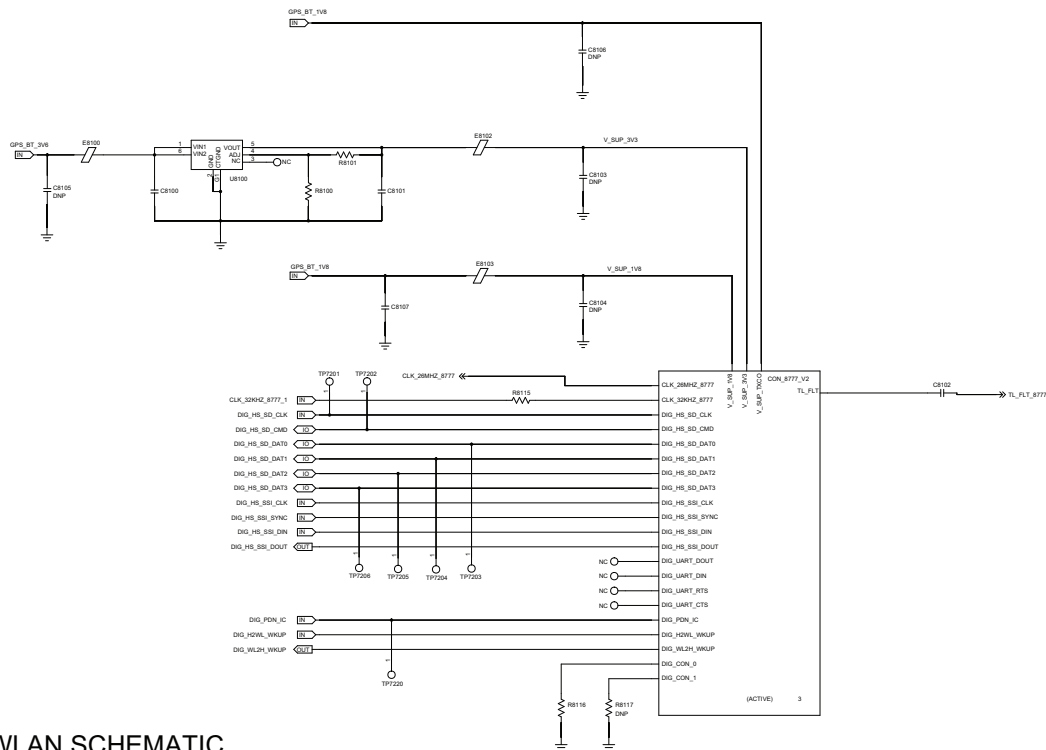
Bluetooth/Wi-Fi/GNSS (PMLN7252_S) Expansion Board – PCB PC000942A01-AA Bottom Side



Bluetooth/Wi-Fi/GNSS (PMLN7252_S) Expansion Board Schematic Diagram (sheet 1 of 4)



Bluetooth/Wi-Fi/GNSS (PMLN7252_S) Expansion Board Schematic Diagram (sheet 2 of 4)





3.0 Bluetooth/Wi-Fi/ GNSS (PMLN7252_S) Expansion Board Parts List – PCB PC000942A01-AA

Circuit Ref.	Motorola Part No.
C7100	21667009370
C7101	21667009174
C7102	2113946S35
C7103	NOT PLACED
C7104	2113946S35
C7106	2113945Y02
C7107	21667009370
C7108	2113946S35
C7109	2113945Y02
C7110	NOT PLACED
C7114	21667009370
C7127	2113945Y02
C7128	NOT PLACED
C7129	21667009355
C7130	21667009355
C7131	21012182001
C7132	21012043035
C7133	2113956A51
C7134	2113956A51
C7135	2113956A51
C7136	2113956A51
C7137	2113956A51
C7138	2170282H06
C7139	2113956A51
C7140	2113956A51

Circuit Ref.	Motorola Part No.
C7141	21667009370
C7142	2113944A52
C8100	2187893N01
C8101	2187893N01
C8102	21667013140
C8103	NOT PLACED
C8104	NOT PLACED
C8105	NOT PLACED
C8106	NOT PLACED
C8107	2113946D05
C8300_3	21009304001
C8305_3	21667012146
C8306_3	21009304001
C8311_3	21009304001
C8312_3	2113946U01
C8315_3	21012129001
C8316_3	21009304001
C8319_3	2113946U01
C8320_3	2113946U01
C8321_3	2113946U01
C8322_3	2113946U01
C8323_3	2113946U01
C8324_3	2113946U01
C8325_3	2113946U01
C8327_3	2113945F07
C8328_3	2113946U01
C8329_3	21013048001
C8331_3	2113946U01
C8332_3	2113946U01
C8334_3	CA000445A01

Circuit Ref.	Motorola Part No.
C8335_3	21667013140
E7100	7688697V14
E8100	7686949J22
E8102	7688697V04
E8103	7688697V04
E8300_3	7688697V14
FL7101	9109674L58
FL7102	91012057001
J7100	09012053001
J7101	0909901V02
J7102	CN000512A01
L7101	24009344027
L7103	2475122C29
L7104	2475122C12
L8301_3	2475122C28
L8330_3	IN000095A01
L8333_3	IN000095A01
R7100	0613952R66
R7105	0613952N01
R7106	0613952N01
R8100	0613952V09
R8101	0613952V30
R8115	0613952R66
R8116	0613952V69
R8117	NOT PLACED
R8300_3	0613952R66
R8301_3	0613952Y66
R8304_3	0613952X73
R8305_3	0613952Y66
R8314_3	0613952Y66

Circuit Ref.	Motorola Part No.
SH7100	2680514Z02
TP8310_3	NOT PLACED
TP8313_3	NOT PLACED
TP8317_3	NOT PLACED
TP8318_3	NOT PLACED
TP8326_3	NOT PLACED
U7103	IC000128A01
U8100	5171734H01
U8300_3	IC000285A01
U8336_3	HZ000243A01
VR7100	4813977A43
Y7101	93012017001
Y8302_3	HZ000202A01
PCB	PC000942A01